SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:		**Date:**	

Code of Conduct

I **D. Naveen / 192372177** *(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

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Annexure A - QUESTIONS

1. Design a concept map that evaluates the significance of the fundamental concepts of mobile computing.
   Establish relationships among mobility, portability, wireless communication, and ubiquitous computing.
   Prioritize the concepts based on their contribution to mobile computing.
   Justify the linking phrases used between the concepts.

2. Construct a concept map that evaluates the effectiveness of the cellular network architecture.
   Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem.
   Identify the most critical components influencing communication.
   Justify your evaluation through meaningful concept relationships.

3. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
   Illustrate their interdependencies.
   Assess which subsystem has the greatest influence on service continuity.
   Justify your conclusions using concept relationships.

4. Develop a concept map that evaluates the effectiveness of the mobile network protocol stack.
   Illustrate interactions among protocol layers.
   Assess how each layer contributes to reliable mobile communication.
   Support your evaluation with meaningful hierarchical relationships.

5. Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN).
   Relate operators, subscribers, switching centers, and radio access networks.
   Assess how these components collectively provide seamless connectivity.
   Justify the relationships included in the concept map.

Rubrics for Calculation

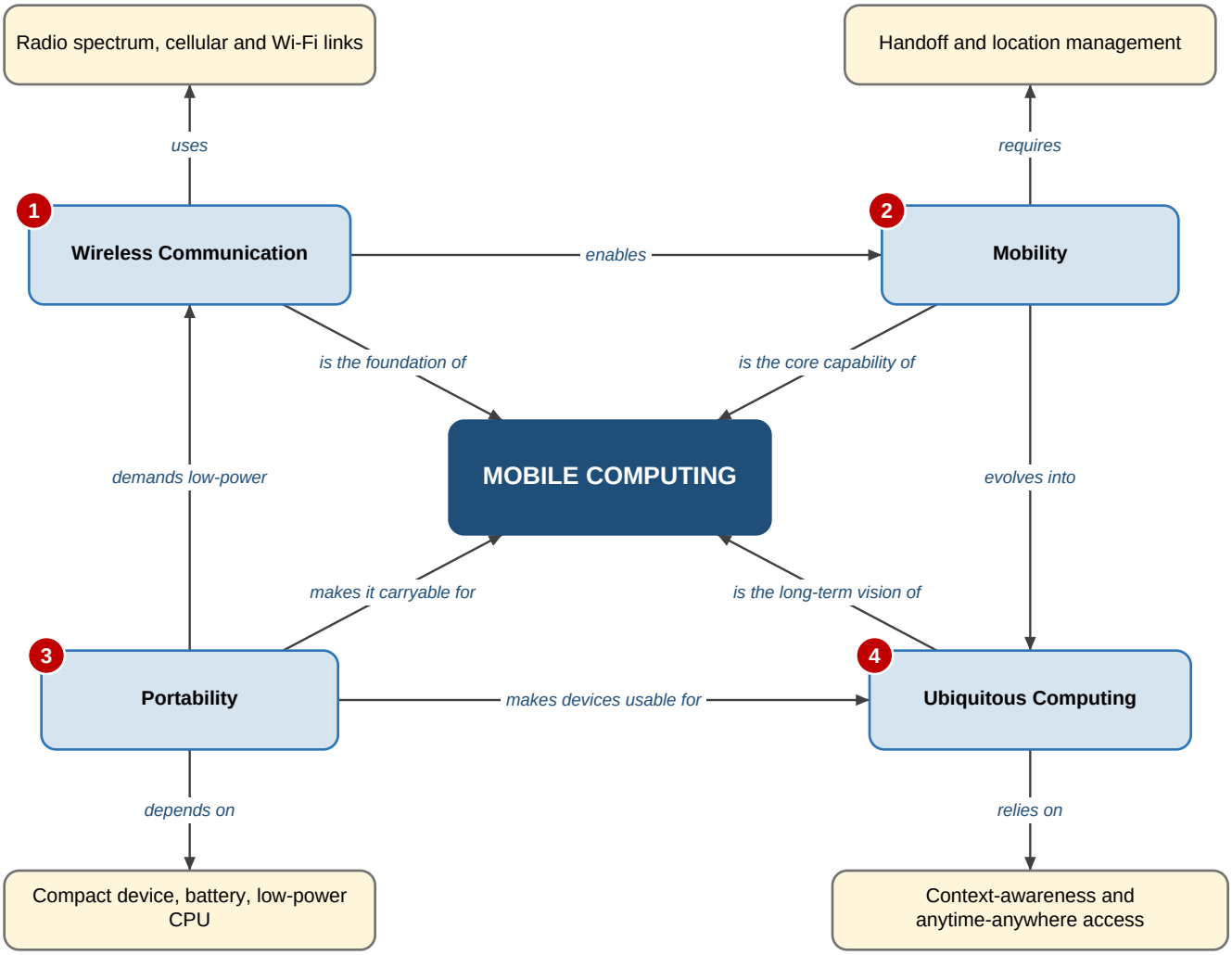
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure B - ANSWER 1

Design a concept map that evaluates the significance of the fundamental concepts of mobile computing. Establish relationships among mobility, portability, wireless communication, and ubiquitous computing. Prioritize the concepts based on their contribution to mobile computing. Justify the linking phrases used between the concepts.



Prioritisation (contribution to mobile computing; rank badge on each concept)

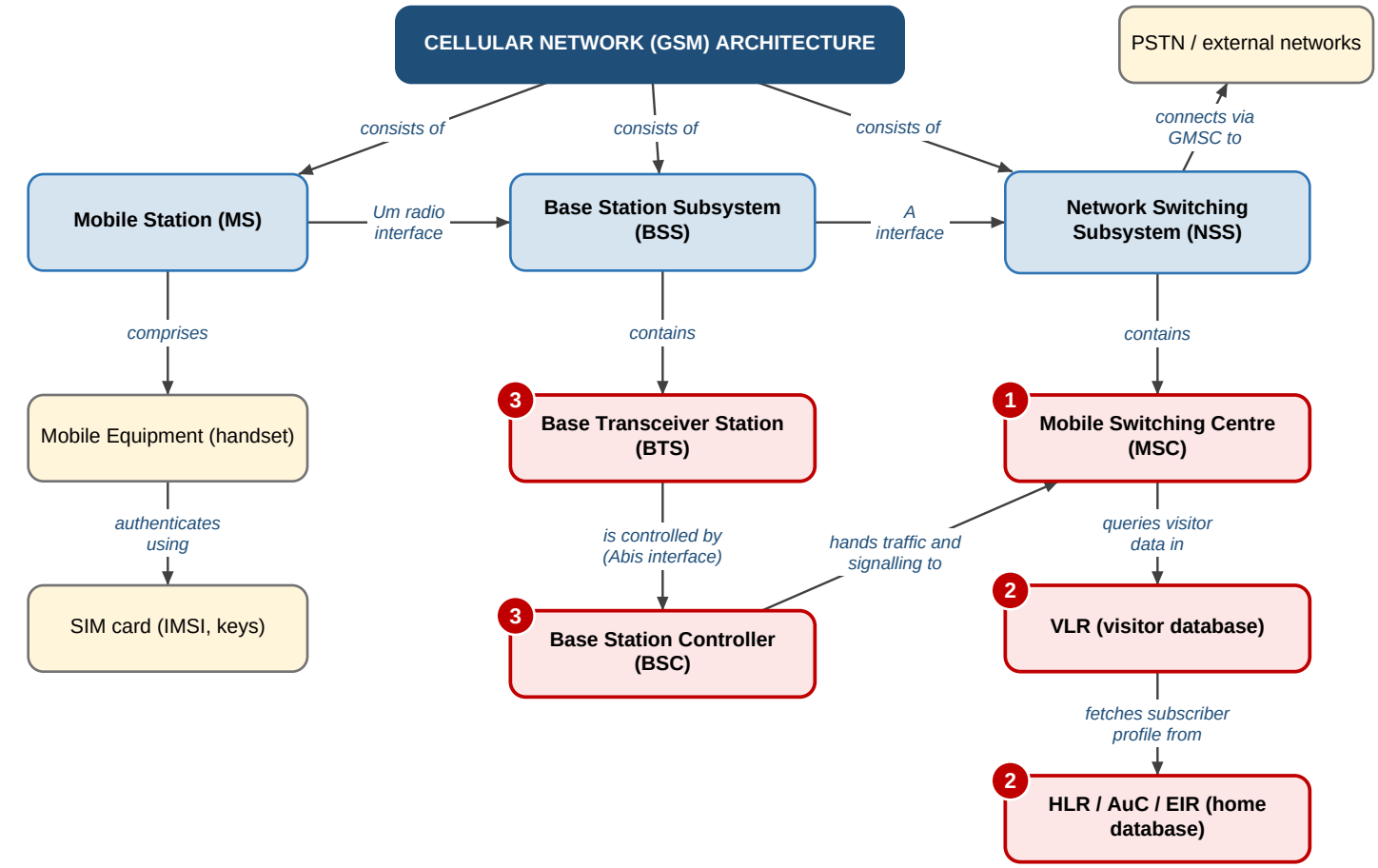
- **1. Wireless communication** - the foundation. Without a radio link no data can move between device and network, so neither mobility nor ubiquity can exist.
- **2. Mobility** - the defining capability: sessions continue while the user changes location (handoff, location management).
- **3. Portability** - the physical enabler: small, light, battery-powered devices that can be carried. A portable laptop used only in an office is portable but not mobile.
- **4. Ubiquitous computing** - the end goal. It appears only when the first three are mature and combined with context-awareness, so it depends on all of them.

Justification of linking phrases

- **"is the foundation of" / "enables"** - wireless is the transmission layer under every mobile service; removing the cable is what allows the user to move.
- **"demands low-power"** - a portable device has a small battery, so the radio and CPU must be energy-efficient; portability therefore constrains wireless design.
- **"evolves into"** - anytime-anywhere access (mobility) plus awareness of context becomes ubiquitous computing; the arrow shows progression, not equality.
- **"makes devices usable for"** - ubiquity needs devices people carry everywhere. **"uses / requires / depends on / relies on"** link each concept to the technique that realises it.
- **Overall:** each arrow reads as a sentence (concept - phrase - concept) and points from the supporting concept to the one it supports, so priority rank follows dependency.

Annexure B - ANSWER 2

Construct a concept map that evaluates the effectiveness of the cellular network architecture. Illustrate the interactions among Mobile Station, Base Station Subsystem, and Network Switching Subsystem. Identify the most critical components influencing communication. Justify your evaluation through meaningful concept relationships.



Red outline + number = criticality rank for communication (1 = most critical).

Evaluation of architectural effectiveness

- **Modular split of duties:** MS = user access, BSS = radio resources, NSS = switching and subscriber intelligence, so each part can be scaled or upgraded independently.
- **Cell-based frequency reuse** multiplies capacity; standard interfaces (Um, Abis, A) allow multi-vendor equipment; central HLR/VLR databases make roaming possible. **Weak point:** MSC and HLR are single points of failure, so they need redundancy.

Most critical components influencing communication

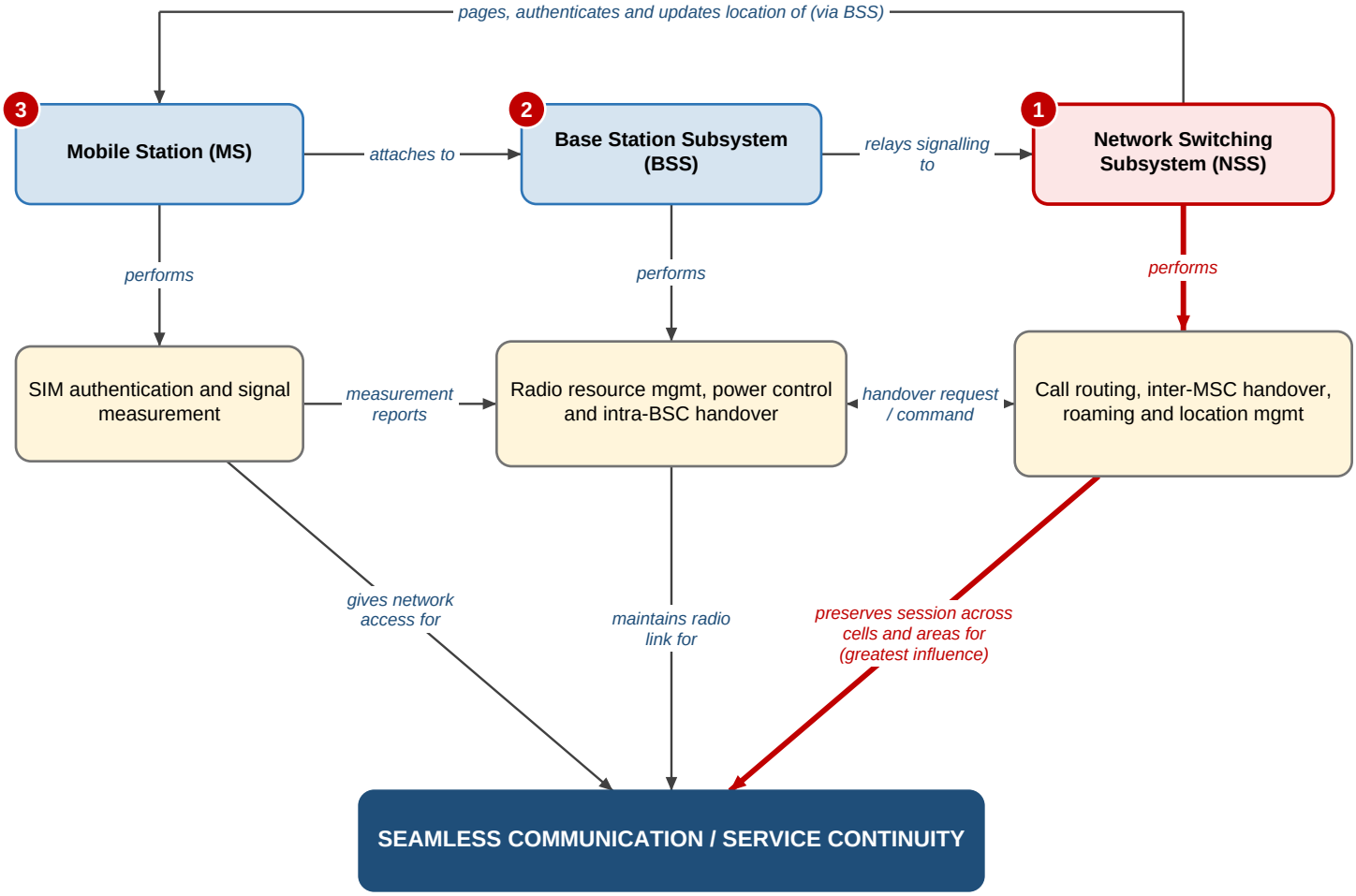
- **1. MSC** - sets up and routes calls, controls handover between BSCs and is the gateway to the PSTN; its failure stops service for the whole area.
- **2. HLR / VLR** - hold subscriber identity, authentication and current location; without them a call cannot be routed or authorised.
- **3. BTS and BSC** - the BTS is the only radio entry point (cell outage = coverage loss); the BSC manages channels and handover for many BTSs. The MS is least critical at system level (one user affected).

Justification of relationships

- **"Um radio interface"** is the only wireless hop; **"A interface"** and **"Abis"** are wired standard interfaces carrying traffic and signalling. **"queries" / "fetches from"** show the VLR-HLR hierarchy used when a subscriber visits a new area. **"connects via GMSC"** shows NSS inter-working with other networks.

Annexure B - ANSWER 3

Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication. Illustrate their interdependencies. Assess which subsystem has the greatest influence on service continuity. Justify your conclusions using concept relationships.



Interdependencies

- The MS depends on the BSS for radio channels; the BSS depends on MS measurement reports for handover and power control, and on the NSS for routing and authentication; the NSS depends on the BSS to reach (page) the subscriber and on the SIM identity from the MS. The dependency is circular, so no subsystem can deliver seamless service alone.

Assessment - which subsystem influences service continuity most

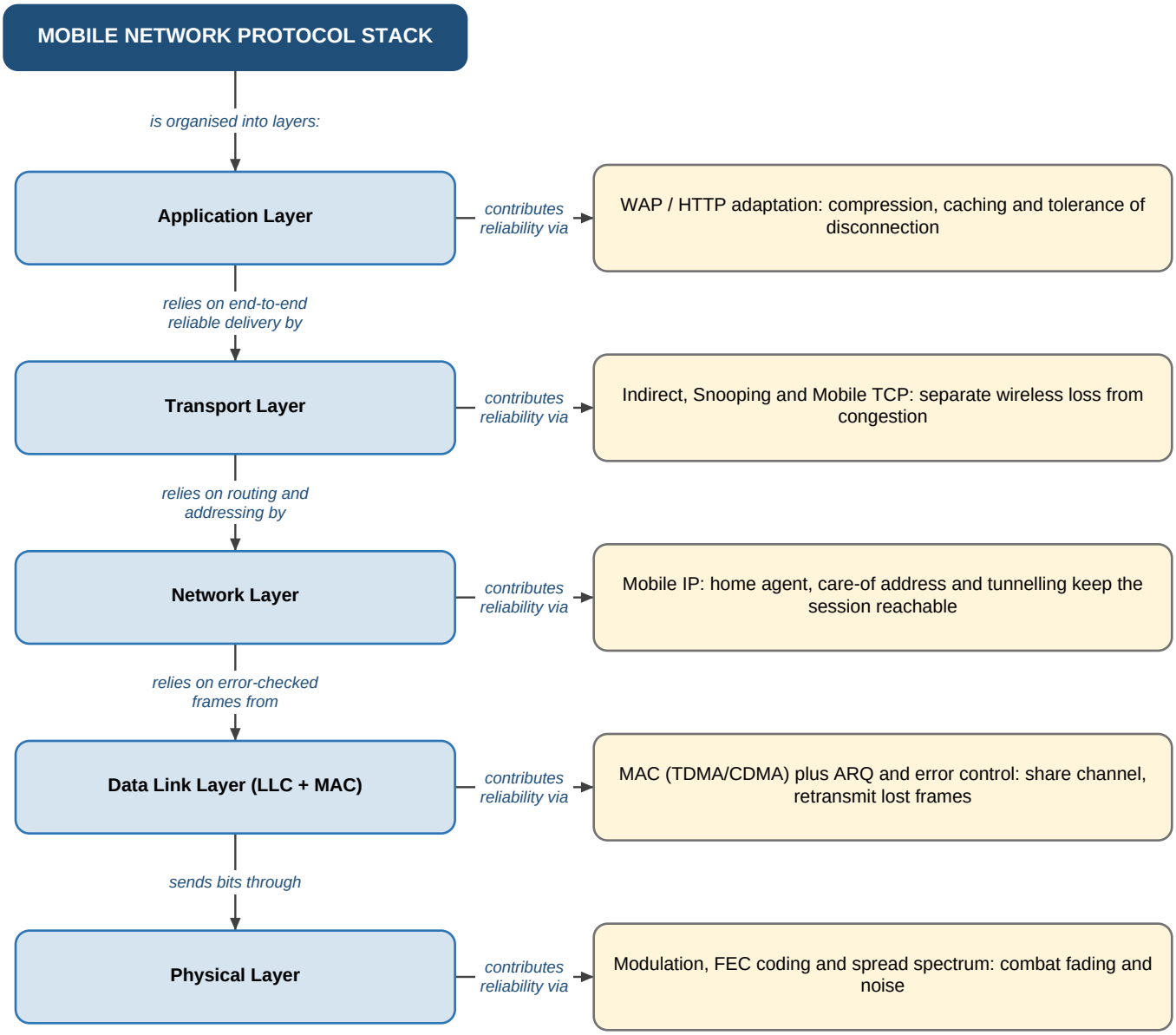
- **1. NSS (greatest):** handles inter-BSC / inter-MSC handover, location updating (HLR/VLR), authentication, roaming and call routing - the functions that keep a session alive across cells, location areas and operators.
- **2. BSS:** executes intra-BSC handover and power control, but a fault only affects the local cells it controls.
- **3. MS:** supplies measurements and identity; continuity depends on it, but a fault affects one user only.

Justification using concept relationships

- **"relays signalling to"** and **"pages, authenticates and updates location of"** close the loop MS-BSS-NSS-MS and prove the interdependence. The two-way **"handover request / command"** link shows the BSS only executes what the NSS decides when the move crosses BSC or MSC boundaries. The thick red link **"preserves session across cells and areas"** marks the NSS as the largest contributor to continuity.

Annexure B - ANSWER 4

Develop a concept map that evaluates the effectiveness of the mobile network protocol stack. Illustrate interactions among protocol layers. Assess how each layer contributes to reliable mobile communication. Support your evaluation with meaningful hierarchical relationships.



Interactions and hierarchy

- The stack is hierarchical: every layer **uses the service of the layer below and offers a service to the layer above**. Data travels down on the sender and up on the receiver; peer layers talk through headers.

Contribution of each layer to reliable mobile communication

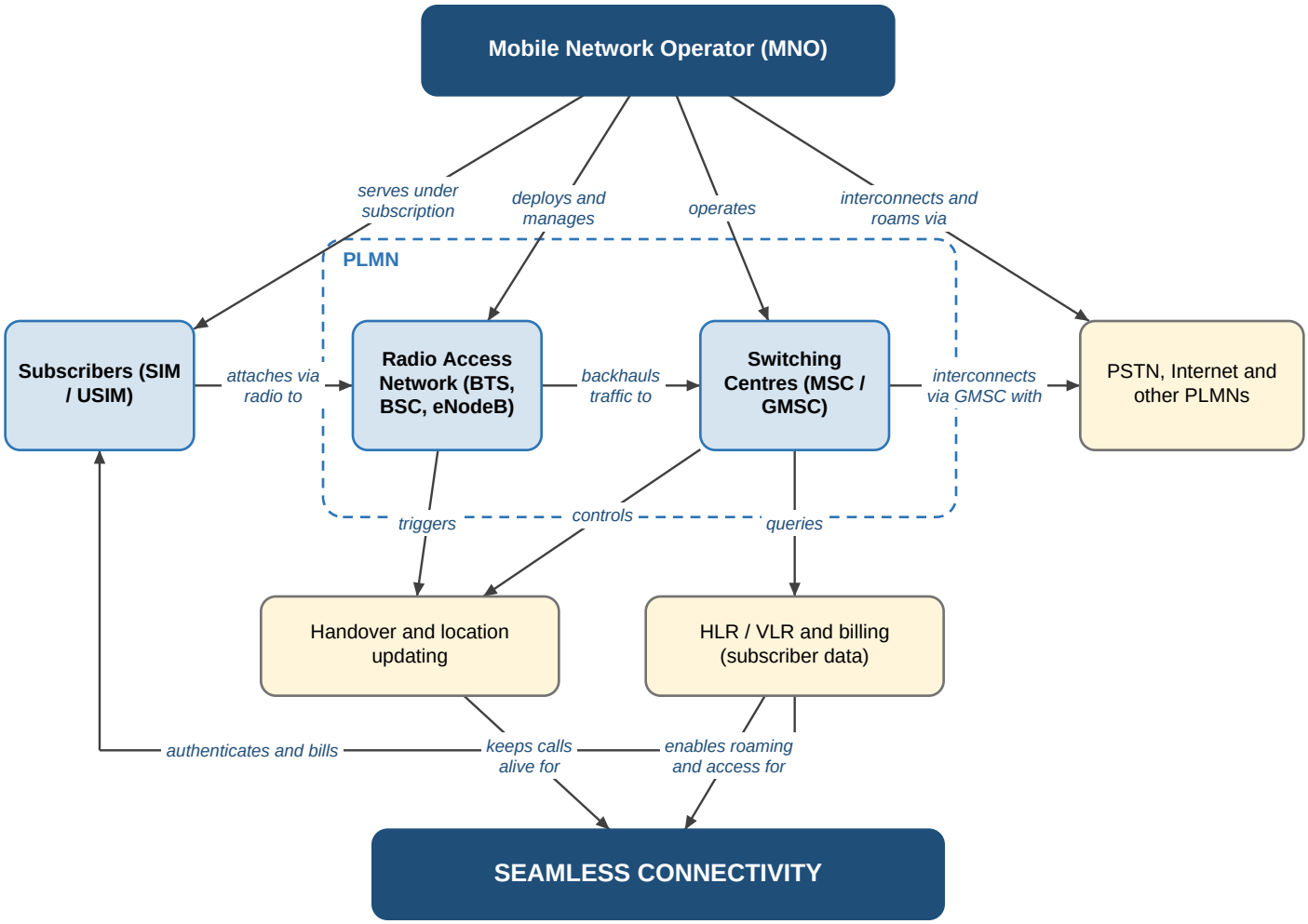
- **Physical:** robust modulation, forward error correction and spread spectrum keep bits intact despite fading, noise and interference.
- **Data link:** MAC arbitrates the shared radio channel (avoids collisions); ARQ retransmits damaged frames and supports handover.
- **Network:** Mobile IP keeps one permanent address (home agent + care-of address + tunnelling) so connections survive movement between cells and networks.
- **Transport:** mobile-aware TCP variants do not treat wireless loss as congestion, so throughput does not collapse. **Application:** WAP / adaptive content copes with low bandwidth and short disconnections.

Evaluation

- The network and transport layers add the most *mobile-specific* reliability (mobility and loss handling) while the physical and data-link layers give the basic link quality; the hierarchy is effective because a fault at a lower layer is masked by recovery at a higher one.

Annexure B - ANSWER 5

Construct a concept map that evaluates the operational efficiency of a Public Land Mobile Network (PLMN). Relate operators, subscribers, switching centers, and radio access networks. Assess how these components collectively provide seamless connectivity. Justify the relationships included in the concept map.



Relationship of the components

- The **operator** owns the PLMN (identified by MCC + MNC), deploys the **RAN** and operates the **switching centres**, and serves **subscribers** under a subscription. Subscribers attach through the RAN; the RAN backhauls traffic to the switching centres; the switching centres consult HLR/VLR/billing and interconnect with PSTN, the Internet and other PLMNs (roaming).

Assessment of operational efficiency and seamless connectivity

- **RAN** gives coverage and capacity through frequency reuse and cell splitting; **switching centres** route calls, control handover and gateway to external networks; **HLR/VLR/billing** keep one subscriber record usable anywhere (authentication, roaming, charging).
- Together, handover and location updating keep sessions alive while the subscriber moves, so connectivity is seamless. Efficiency can be judged by call set-up success, handover success and call-drop rates; the main risks are backhaul and core single points of failure.

Justification of included relationships

- **"attaches via radio to" / "backhauls traffic to"** trace the end-to-end user path. **"queries" / "authenticates and bills"** show why subscriber data must be centralised. **"deploys and manages / operates"** show operator responsibility. **"triggers" / "controls"** show the RAN-core cooperation behind handover. **"interconnects and roams via"** shows the PLMN is not isolated.